

MELILLA, Spain

WMO: 603380

Lat: 35.280N

Lon: 2.956W

Elev: 48

StdP: 100.76

Time Zone: 1.00 (EUC)

Period: 82-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	7.7	8.7	0.4	3.9	12.5	2.1	4.4	13.2	12.9	11.8	11.2	13.4	2.0	250	0.381

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	6.0	32.3	20.1	30.8	20.7	29.2	21.4	24.8	28.0	24.3	27.3	23.8	26.8	4.2	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
23.9	18.8	26.8	23.3	18.1	26.4	22.9	17.7	26.1	75.6	28.0	73.4	27.3	71.5	26.8	27.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.6	9.2	8.0	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	19.2	13.7	14.2	15.5	16.8	19.4	22.4	25.5	26.2	23.7	20.7	17.2	14.8		
	DBStd	4.68	1.92	1.99	1.90	1.67	1.94	1.85	1.88	1.84	1.89	2.00	1.95	2.01		
	HDD10.0	2	1	1	1	0	0	0	0	0	0	0	0	0		
	HDD18.3	573	145	117	89	52	11	1	0	0	0	5	44	110		
	CDD10.0	3361	114	117	172	204	291	371	479	502	412	331	217	150		
	CDD18.3	891	1	0	3	6	43	122	221	243	162	77	10	2		
	CDH23.3	5577	2	0	14	18	109	484	1769	2139	857	168	11	6		
	CDH26.7	1332	0	0	1	1	10	73	502	606	123	14	0	2		
(14) Wind	WSAvg	3.5	3.3	3.8	4.0	3.9	3.8	3.8	3.4	3.5	3.2	2.9	3.2	3.3		
(15) Precipitation	PrecAvg	384	59	48	44	36	21	7	1	3	19	40	55	51		
	PrecMax	761	167	118	164	185	62	107	8	25	131	261	189	160		
	PrecMin	217	0	2	3	6	3	0	0	0	0	1	2	0		
	PrecStd	120	45	33	41	38	17	22	2	6	26	52	45	38		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	21.9	22.0	25.5	25.7	28.0	31.1	37.0	34.8	31.2	28.1	24.5	23.5		
		MCWB	14.1	15.1	15.2	17.0	17.5	21.3	19.0	19.9	20.8	19.1	16.4	15.4		
	2%	DB	19.3	20.0	22.0	23.1	26.0	28.2	32.3	32.8	29.2	26.2	23.0	20.9		
		MCWB	13.8	14.6	14.7	15.8	17.5	20.2	19.9	20.5	20.8	19.7	16.4	14.9		
	5%	DB	18.0	18.6	20.3	21.5	24.4	26.9	30.6	31.1	28.1	25.1	21.8	19.1		
		MCWB	13.1	14.0	14.7	15.5	17.4	20.1	20.3	21.1	21.3	20.0	16.7	14.7		
	10%	DB	16.8	17.5	19.0	20.2	23.2	25.8	29.0	29.6	27.0	24.1	20.7	17.9		
		MCWB	12.7	13.6	14.5	15.1	17.1	19.8	21.2	21.8	21.1	19.6	16.0	14.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	16.8	16.9	17.4	18.9	20.0	23.6	24.7	25.7	24.7	22.2	19.3	17.5		
		MCDB	20.1	19.9	20.9	23.2	23.7	26.7	28.6	29.4	27.2	24.5	22.2	21.0		
	2%	WB	15.0	15.5	16.3	17.6	19.1	22.4	24.1	25.1	24.1	21.5	18.3	16.2		
		MCDB	17.8	18.5	19.7	21.0	23.1	25.7	27.6	28.4	26.5	24.1	21.1	18.8		
	5%	WB	14.0	14.7	15.6	16.7	18.5	21.6	23.7	24.7	23.6	20.9	17.4	15.4		
		MCDB	16.8	17.6	18.8	19.9	22.5	25.0	27.1	27.8	26.1	23.7	20.1	18.0		
	10%	WB	13.3	14.0	15.0	15.9	18.0	21.0	23.2	24.2	23.0	20.4	16.7	14.8		
		MCDB	16.0	16.8	17.9	18.7	21.8	24.4	26.5	27.2	25.7	23.0	19.3	17.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	6.5	6.3	6.2	6.6	6.5	6.3	6.6	6.0	6.0	6.2	6.4	5.9		
		MCDBR	7.9	8.0	9.0	9.0	8.7	7.9	8.9	8.4	7.6	7.5	7.9	7.5		
	5% WB	MCWBR	5.0	5.1	5.3	5.2	4.5	4.2	4.9	5.3	4.0	4.2	4.5	4.4		
		MCWBR	6.8	6.9	7.3	7.5	7.3	6.5	6.4	6.3	5.6	6.3	6.5	6.2		
(40) Clear-Sky Solar Irradiance	taub		0.327	0.355	0.409	0.417	0.446	0.480	0.516	0.504	0.462	0.415	0.365	0.334		
	taud		2.515	2.438	2.265	2.268	2.210	2.116	2.016	2.062	2.194	2.319	2.448	2.514		
	Ebn at Noon		876	885	859	870	845	813	780	781	796	805	820	843		
	Edh at Noon		82	99	128	134	144	158	173	163	136	110	88	78		
(44) All-Sky Solar Radiation	RadAvg		2.78	3.64	4.78	5.82	6.69	7.42	7.25	6.46	5.25	3.97	2.92	2.52		
	RadStd		0.22	0.31	0.42	0.43	0.54	0.24	0.20	0.26	0.23	0.25	0.22	0.18		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (6 neighbors)		+0.25	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page